SPATIAL LOCATION IN THE LEARNING OF A MULTIPLE-T MAZE

GEORGE GENTRY, W. LYNN BROWN AND HUN LEE

Departments of Philosophy and Psychology, The University of Texas[1]

Received February 19, 1948

In a recent study by Gentry, Brown, and Kaplan (1) on spatial learning, results were obtained which were at variance with results found by Tolman, Ritchie, and Kalish (2). Both studies gave the animals four periods of preliminary training on the same type of apparatus and used the same type of apparatus on the test trial. Tolman, Ritchie, and Kalish found that 36 per cent of their animals chose the short-cut path from eighteen novel available pathways when the tunnel path (CD) was blocked on the test trial. The tunnel path was used by the animals during preliminary training. They assert that the animals "were running toward the location of the former goal, and that this location was indicated by the position of the light (p. 21)." Gentry, Brown, and Kaplan found, in a repetition of the work by Tolman, Ritchie, and Kalish, while using four periods of training, that their animals had a tendency (37.5 per cent) to choose pathways adjacent to the blocked alley, and that none chose the short-cut path to the former location of the food box.

Tolman, Ritchie, and Kalish assume that the 64 per cent of their animals which did not choose the short-cut on the test trial had not been given sufficient training to determine the location of the food box. We quote:

> We believe that the reason that the remaining rats failed to take the shortcut was that they had not had enough training and thus had not yet learned the location of the food. With a few more days training we should have expected that the remaining rats would have chosen the short-cut (pp. 21–22).

Gentry, Brown, and Kaplan failed to get a tendency to choose the short-cut path on the test trial following nine periods of training, rather than four, as was the case in the study by Tolman, Ritchie, and Kalish. Gentry, Brown and Kaplan found that 30 per cent of their animals chose paths 9 and 10, paths adjacent to the blocked alley, on the test trial following nine periods of training.

Gentry, Brown and Kaplan explained their results in part in terms of an artifact of the experimental apparatus, *i.e.*, the presence or absence of blocked alley CD on the test trial. It appears to us that the explanation by Tolman, Ritchie, and Kalish of their results involves the assumption that their rats possessed a capacity to discriminate between the eighteen novel available pathways, without distinguishable features at the choice points, with which they were confronted in the test trial, and to determine (infer) without error that No. 6 pathway was a short-cut, *i.e.*, the pathway to location L. Gentry, Brown, and Kaplan "found no evidence that any of the rats . . . used had such a capacity and much evidence that none of them did (p. 321)."

[1] The cost of this study was met in part by a grant to the authors by the Research Institute of The University of Texas.

The studies by Tolman, Ritchie, and Kalish (2) and Gentry, Brown, and Kaplan (1) did not have a criterion of learning during the preliminary training period. There was no basis of error in performance or criterion of proficiency in performance. The present study was designed to test spatial learning on a test trial following the *learning* of a multiple-T maze.

MATERIALS AND METHODS

Animals: Eighty animals were used in the present study. The group was composed both of animals purchased from the Wistar Institute and albinos and hooded animals reared in the Comparative Psychology Laboratory at The University of Texas. They were be-

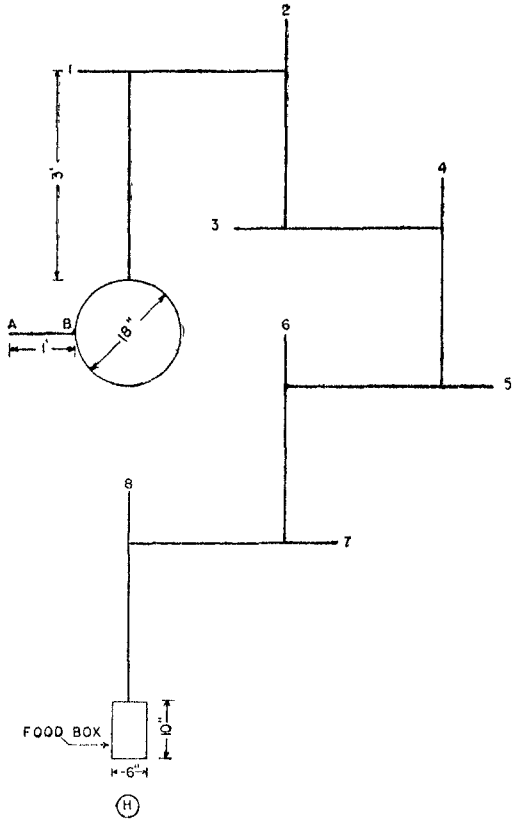


FIG. 1. MAZE USED IN PRELIMINARY TRAINING

tween 100 and 150 days old at the beginning of the experiment. They had had no previous maze experience. All training and the test trial were at night between seven and eleven P.M.

Experimental room: The experimental room was eighteen feet square with a window on the north, east, and west, and a door on the south. Two chairs and the experimental apparatus were the only equipment in the room during the experiment.

Apparatus: The maze used in preliminary training prior to the test trial was an elevated eight-unit multiple-T maze. Figure 1 presents in diagram, with the dimensions listed, the maze used in the preliminary training period. The maze was taken apart daily and the

units washed. The units were shifted in the maze pattern from day to day. The maze was assembled each night in the same place in the room. The units were 2″ by 36″ and were of pine and painted black. The eighteen-inch starting table and unit AB were also painted black. An individual food box 6″ x 6″ x 10″ was used for each animal. A 7.5 watt bulb in a desk lamp at H was the only illumination in the room. The reflector of the desk lamp was turned so that the light would shine directly down pathway eight.

In figure 2 we present in diagram, with the dimensions listed, the apparatus used in the test trial. The light, starting section AB, the table and section one of the maze (blocked pathway on test trial) remained in the same position in the room in the test trial which

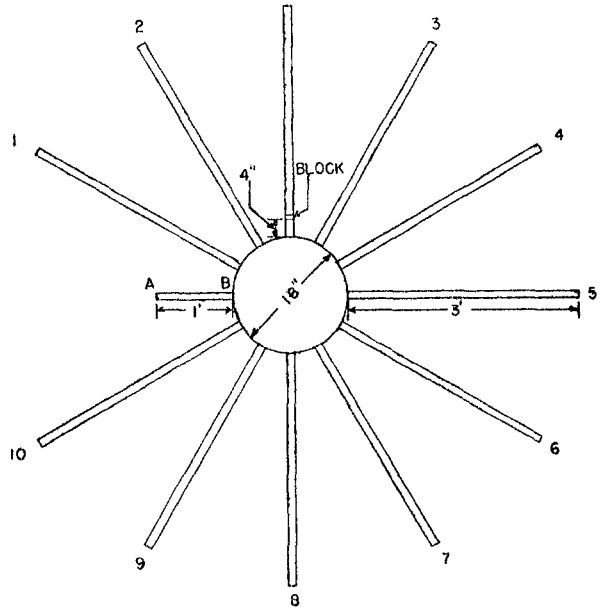


FIG. 2. APPARATUS USED IN TEST TRIAL

they occupied during preliminary training. The food box was removed on the test trial. The table could be rotated independently of the pathways.

Scoring: Only one kind of error was recorded: body length entrance into a cul. Only one error per section was recorded for each run. Retracing errors were not used in presenting results.

PROCEDURE AND RESULTS

Preliminary training: The animals were given one trial a day on the apparatus shown in figure 1 between seven and eleven P.M. for eleven days. They were permitted to eat for about five minutes in individual food boxes after each trial before being returned to their home cages where they received additional food. Purina Dog Chow was used in the reward box and in the home cages.

Results of preliminary training: We present the results of the preliminary training period in figure 3 in terms of error scores. An examination of these data will show that the animals had learned the maze by the eleventh trial, the average error per animal per run being .35 in an eight cul maze. It was as-

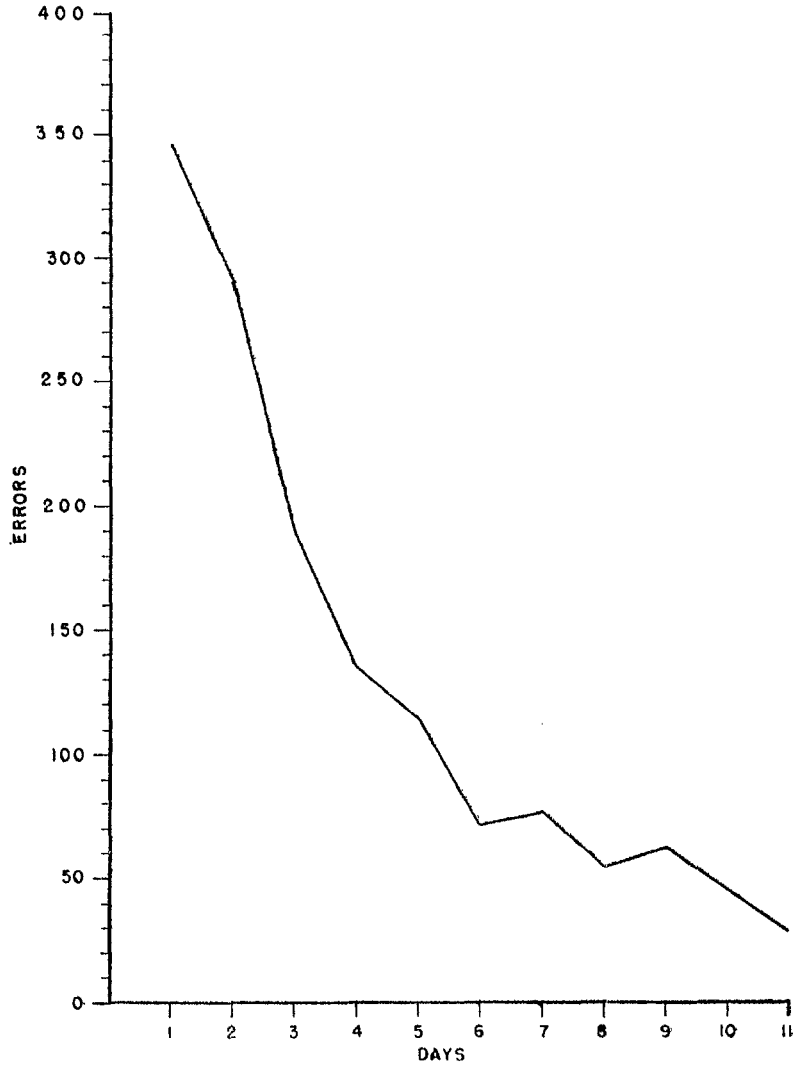


FIG. 3. ERROR SCORE CURVE

sumed that when the animals reached this low error score they had learned the maze and were ready for the test trial.

Test procedures: The apparatus used in the test trial is shown in diagram, with dimensions listed, in figure 2. The test trial was at night between seven and eleven P.M. The former true path in the pretest training was blocked

four inches from the starting table. Each rat was placed at A and allowed to explore the apparatus and enter the blocked pathway before making the critical choice. The pathway taken by the animal after it had entered the blocked pathway was scored as the selection in the test trial. The animal's test trial ended as soon as it ran to the end of the selected pathway following entrance into the blocked pathway. The starting table was rotated frequently during the test period in order to prevent any possibility of "tracking." Most animals

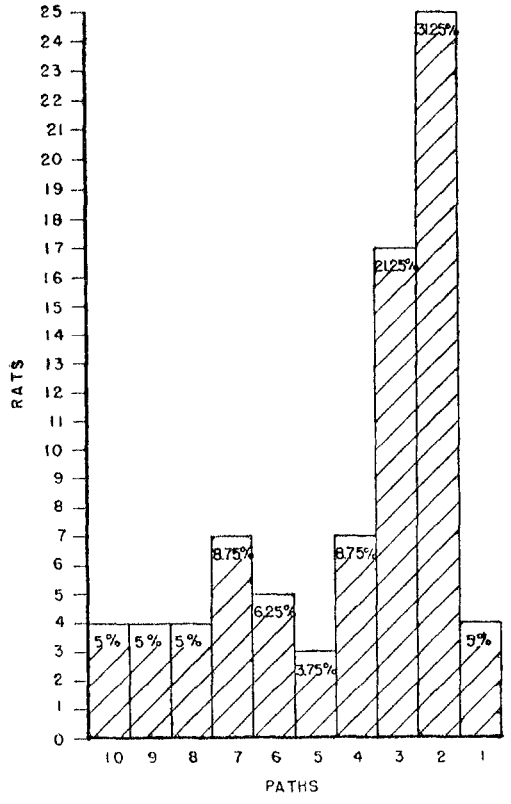


FIG. 4. NUMBER OF RATS WHICH CHOSE EACH OF THE PATHS

went immediately to the blocked pathway as soon as they reached the table from section AB. Many animals investigated the block in the pathway several seconds before running one of the ten available pathways. The animals showed no hesitation in running the pathways from the starting table in the test trial.

Results of the test trial: We present the results of the test trial in figure 4. These data show that 52.5 per cent of the animals chose paths 2 and 3, paths adjacent to the blocked path, in the test trial, and 5 per cent chose path 8, the pathway leading directly toward the light at H and the position of the food box in preliminary training. The data in figure 4 show that the animals did not have any preferences between the remaining eight pathways.

DISCUSSION OF RESULTS AND CONCLUSIONS

The findings of this experiment appear to support both the negative and positive interpretations suggested by Gentry, Brown, and Kaplan in their discussion of the results obtained in their repetition of the first of the Tolman, Ritchie, and Kalish "Studies in spatial learning.—Orientation and the short-cut." Although the conditions during the preliminary training were more favorable for learning the location of the goalbox and for developing a directional orientation with reference to it, we found no evidence of either. The behavior of the animals on the test trial gave us no reason for believing that the light (or locus of the goalbox) had any material bearing on the performance of the animals or reasons for thinking that they had acquired even the rudiments of a general directional orientation with reference to it (or them). Moreover, there was nothing about their behavior when confronted with the problem of choosing the short-cut from among the ten available pathways that we observed that suggested that the animals could have accomplished this feat even if they had learned the locus of the goal-box and/or light and had acquired the appropriate orientation.

The results show that the test trial choices of the eighty subjects fall roughly into two groups: (a) those concentrated on the two pathways immediately adjacent to the blocked pathway, two and three, and (b) those distributed more or less equally among the remaining eight pathways. The animals exhibited a tendency either to commit themselves to a pathway in the immediate proximity of the true pathway during the training runs or to wander about and choose more or less indiscriminately among the remaining possibilities.

The fact that 52.5 per cent of the total choices were of pathways two and three is of particular interest not merely in that they were choices of pathways immediately adjacent to the blocked pathway but in that they were in approximate directional opposition to the light, the locus of the goalbox, and the short-cut pathway. This tendency, even more pronounced than that exhibited by the Gentry, Brown, and Kaplan findings, to choose pathways simulating the original true pathway in point of departure and direction of departure from the table top is in all probability a direct function of the training. The animals probably acquired a directional orientation and tendency to run in a given direction from a given part of the table top or learned that the appropriate point of departure from the table top was a given area and that progress in a given direction from this area of departure was a step in the action sequence leading to the goalbox. When confronted with the block in the original true pathway this directional orientation, running tendency, etc., maintained itself and tipped the scales in favor of the pathways most closely simulating the original true pathway in point of departure and direction of departure from the table top.

The records show that all members of the remaining group chose some pathway and avoided the pair adjacent to the blocked pathway. But the records show no evidence of a positive preferential choice trend. The behavior of individual members of this group suggested that final committment to a given pathway

after finding the original true pathway blocked was one of the chance variety, or a function of influences that escaped observation. It is possible that such animals, even though they learned the true route from the table top to the goalbox, the correct point of departure, direction of departure, etc., and certainly acquired a disposition to run a pathway leading away from the table top, failed to acquire a directional orientation, or a tendency to run in a given direction, and from a given area, of sufficient strength to persist in presence of the blocked pathway and eventually to determine the choice.

The animals in this experiment exhibited no evidence that the light was a factor or that a general directional orientation with reference to the locus of the goal box was operative. The writers are of the strong opinion that their findings and those of Gentry, Brown, and Kaplan contribute a body of evidence casting serious doubt on the validity of the assumptions underlying the spatial location hypothesis.

REFERENCES

1. Gentry, George, Brown, W. Lynn, and Kaplan, Sylvan J.: An experimental analysis of the spatial location hypothesis in learning. *J. comp. physiol. Psychol.*, 1947, **40**, 309–322.
2. Tolman, E. C., Ritchie, B. F., and Kalish, D.: Studies in spatial learning. I. Orientation and the short-cut. *J. exper. Psychol.*, 1946, **36**, 13–24.